

1 Group theory

The following sections are part of the book *Mathematical methods for physicists* by G. Arfken and H. Weber, chapter 4 and were taken literally word by word. The book goes more into detail on certain points so consider reading the book directly. All information that was added in this section originates from Wikipedia.

1.1 Definition of a Group

A group G may be defined as a set of objects or operations, rotations, transformations, called the elements of G , that may be combined, or “multiplied,” to form a well-defined product in G , denoted by a $*$, that satisfies the following four conditions.

1. If a and b are any two elements of G , then the product $a * b$ is also an element of G , where b acts before a ; or $(a, b) \rightarrow a * b$ associates (or maps) an element $a * b$ of G with the pair (a, b) of elements of G . This property is known as “ G is closed under multiplication of its own elements.”¹
2. This multiplication is associative: $(a * b) * c = a * (b * c)$.
3. There is a unit element 1 in G such that $1 * a = a * 1 = a$ for every element a in G . The unit is unique: $1 = 1' * 1 = 1'$.
4. There is an inverse, or reciprocal, of each element a of G , labeled a^{-1} , such that $a * a^{-1} = a^{-1} * a = 1$. The inverse is unique: If a^{-1} and a'^{-1} are both inverses of a , then $a'^{-1} = a'^{-1} * (a * a'^{-1}) = (a'^{-1} * a) * a^{-1} = a^{-1}$.

Since the $*$ for multiplication is tedious to write, it is customary to drop it and simply let it be understood. From now on, we write ab instead of $a * b$.

- If a subset G' of G is closed under multiplication, it is a group and called a **subgroup** of G ; that is, G' is closed under the multiplication of G . The unit of G always forms a subgroup of G .
- If $gg'g^{-1}$ is an element of G' for any g of G and g' of G' , then G' is called an **invariant subgroup** of G . The subgroup consisting of the unit is invariant. If the group elements are square matrices, then $gg'g^{-1}$ corresponds to a similarity transformation (see ??).
- If $ab = ba$ for all a, b of G , the group is called **commutative** or **abelian**, that is, the order in products does not matter; commutative multiplication is often denoted by a $+$ sign. Examples are vector spaces whose unit is the zero vector and $-a$ is the inverse of a for all elements a in G .

Another few useful definitions (taken from Wikipedia) are as follows.

- A group G is called **cyclic**, if it can be generated by repeated application of one of its elements, i.e. $G = \{g^n \mid n \in \mathbb{Z}\}$ with $g \in G$.
- The **order** of a group G denotes the number of elements $|G|$ of the group.

¹This means that the group operation is defined as $*$: $G \times G \rightarrow G$ with $(a, b) \mapsto a * b$.

- A **generating set of a group** is a subset of the group set such that every element of the group can be expressed as a combination (under the group operation) of finitely many elements of the subset and their inverses.
- The element g of a cyclic group $G = \{g^n \mid n \in \mathbb{Z}\}$ is the **generator** of that group.

1.2 Homomorphism and isomorphism

There may be a correspondence between the elements of two groups: one-to-one, two-to-one, or many-to-one. If this correspondence preserves the group multiplication, we say that the two groups are **homomorphic**. If the correspondence is one-to-one, still preserving the group multiplication,² then the groups are **isomorphic**.

If a group G is homomorphic to a group of matrices G' , then G' is called a **representation** of G . If G and G' are isomorphic, the representation is called **faithful**. There are many representations of groups; they are not unique.

Example: Rotations The set of counterclockwise coordinate rotations of three-dimensional Euclidean space about its z -axis is an example of a group. If the z -axis is fixed, the linear transformation is defined through an angle φ of the xy -coordinate system to a new orientation³. One can write these rotations for example with matrices $R_z(\varphi)$ as

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = R_z(\varphi) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (1)$$

One can easily show, that $R_z(\varphi_1)R_z(\varphi_2) = R_z(\varphi_1 + \varphi_2)$ and thereby see, that the matrices $R_z(\varphi)$ are a representation of the group of rotations. This group of rotation matrices is **abelian**, since the angles of successive rotations simply get added up. Striking off the coordinate vectors in eq. (1), we can associate the matrix of the linear transformation with each rotation, which is a group multiplication preserving one-to-one mapping, an isomorphism: The matrices form a faithful representation of the rotation group.

The unity with angle $\varphi = 0$ and the rotation with $\varphi = \pi$ form a finite subgroup. The finite subgroups with angles $2\pi m/n$, n an integer and $m = 0, 1, \dots, n-1$ are **cyclic**; that is, the rotations $R(2\pi m/n) = R(2\pi/n)^m$ and thus $R(2\pi/n)$ is the generator of the finite subgroup.

Example: Permutations

²Suppose the elements of one group are labeled g_i , the elements of the second group h_i . Then $g_i \leftrightarrow h_i$ is a one-to-one correspondence for all values of i . If $g_i g_j = g_k$ and $h_i h_j = h_k$, then g_k and h_k must be the corresponding group elements.

³Applying two arbitrary rotations by φ_1 and φ_2 successively yields one rotation by $\varphi_1 + \varphi_2$. Thus, the group of rotations is actually closed under multiplication of its own elements. Furthermore, the inverse of φ is simply $\varphi^{-1} = -\varphi$ and the unit $\mathbb{1}$ is given by the rotation with angle $\varphi = 0$.

1.3 Reducible and irreducible representations

Representation: Some other way of describing the same group. Example: matrix representation as shown for rotations in $3d$ space around a fixed axis can be written with matrices. $\rightarrow$ important in physics since many things are represented as vectors.